Schedule to Section 63 of the Bharatiya Sakshya Adhiniyam, 2023

Part B — Technical Expert Certification (Section 79A, IT Act authorised expert)

Expert Name & Designation	Dr. Authorized Forensic Examiner — Digital Forensic Examiner (Sec 79A IT Act)
Expert ID / Licence	IT-79A-IND-2026-XXXX
Verification Workstation Serial	DEMO-WS-2026-0001
Verification MAC	00:11:22:33:44:55
Independent SHA-256 Confirmation	`b94d27b9934d3e08a52e52d7da7dae55cfe5f806fb7c9` (verified match)
Date & Time of Examination (IST)	02-06-2026 10:40:34 IST

Forensic Analysis Summary

The electronic record identified above was extracted using write-protected forensic procedures. Structural integrity of the bitstream was preserved. No transcoding or lossy re-encoding was applied during acquisition.

- Spatial CNN manipulation probability: **0.9200**

- 3D face mesh landmark variance: **0.075000**

- Lip-sync alignment error: **135.00 ms**

- TTS synthetic voice probability: **0.8800**

- Spectrogram pitch mismatch ratio: **0.3900**

- Anti-spoofing NN confidence (bona fide): **0.1100**

Resolved Subject: Shri Demo Politician (ID: candidate-042, fused similarity: 1.0000) — source: hybrid

Grounds for Deepfake Classification

Summary: Media classified as likely synthetically manipulated based on 11 forensic indicator(s), including 4 high-severity finding(s).

- **Audio-Video Sync (critical):** Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech. *(lip_sync_alignment_error_ms: 135.0 ms)*

- **Facial Manipulation (high):** Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation. *(spatial_cnn_manipulation_probability: 92.00%)*

- **3D Face Geometry (medium):** 3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces. *(face_mesh_landmark_variance: 0.0750)*

- **Synthetic Voice (high):** The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording. *(tts_synthetic_probability: 88.00%)*

- **Voice Authenticity (medium):** Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing. *(spectrogram_pitch_mismatch_ratio: 39.00%)*

- **Anti-Spoofing (high):** Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed. *(anti_spoofing_nn_confidence: 11.00%)*

- **Lighting Consistency (medium):** Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture. *(illumination_inconsistency_score: 71.00%)*

- **Shadow Geometry (medium):** Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos. *(shadow_geometry_inconsistency_score: 66.00%)*

- **Color Grading (medium):** Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video. *(color_grading_anomaly_score: 59.00%)*

- **Temporal Stability (medium):** High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation.
  (temporal_flicker_score: 55.00%)

- **Facial Boundaries (medium):** Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body.
  (facial_boundary_artifact_score: 62.00%)

I certify that the above analysis was conducted on the hash-verified original and that the opinions expressed are based on validated multi-modal deepfake detection models.

Expert Signature: ___________________________ Date: 02-06-2026